①

# Pipelining and Superscalar Architecture

Architecture → like a parallel system.

Architecture → Parallel system.

1

→ Instructions → Control unit → ALU ↔ Memory

2

Inst 1 → CU → Process 1

Inst 2 → CU → Process 2

⋮

Inst n → CU → Process n.

⇒ After one Instruction completion, the next Instruction will fetch.

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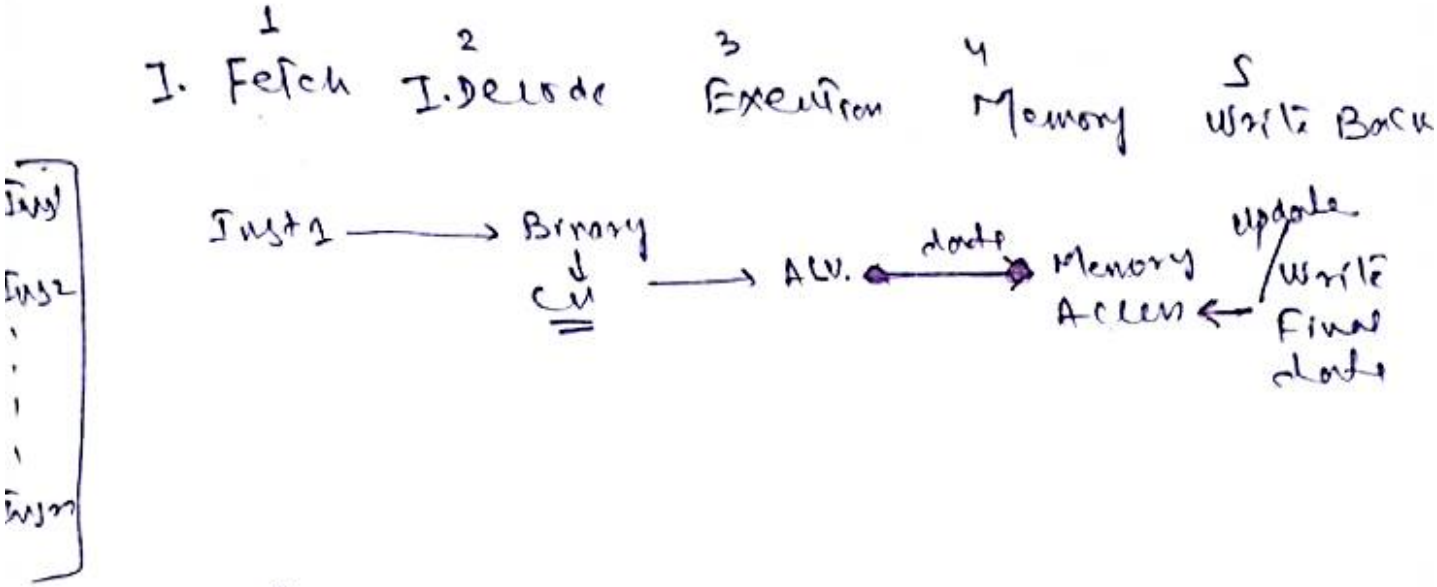
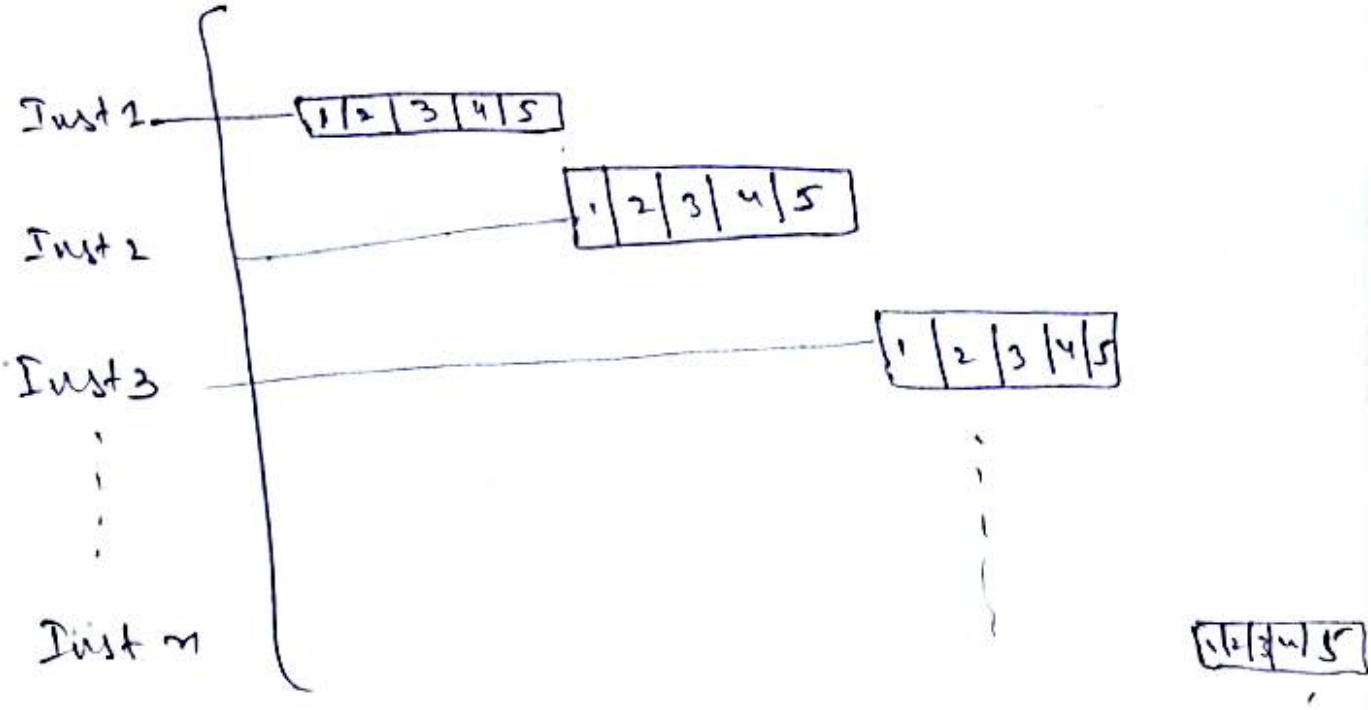
(2) e.g assume. 5-stage Architecture




Key notes

1. Inst 1 Time = Inst 2 Time = ... Inst n Time

Single cycle Architecture.

→ Same Processing | completion time for all type of Instruction.

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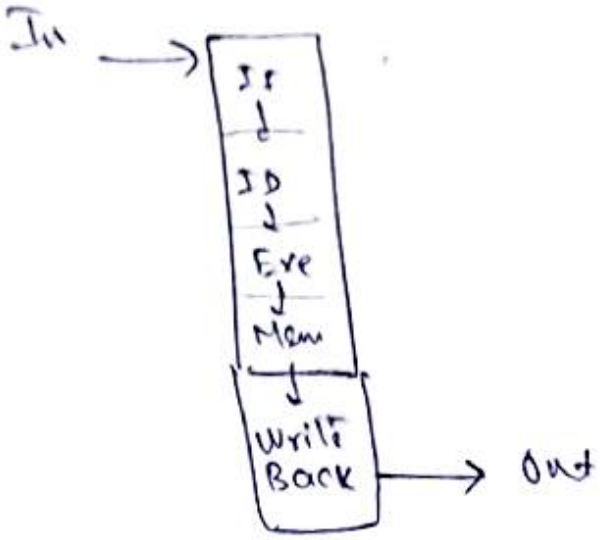
(3)




I → Point for In

J → Point for Out.

Single cycle Arch

→ Focus on <u>worst</u> Instruction.

↓

Longest

Longest Inst Time = clock cycle time

e.g →

Inst 1 = 4 n.s
Inst 2 = 3 n.s
Inst 3 = 4 n.s
Inst 4 = 5 n.s
Inst 5 = 3 n.s

Single cycle ⇒ Focus on longest = 5 n.s

So $\boxed{\text{clock Time} = 5 \text{ n.s}}$

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Inst 1 = 5 n.s
Inst 2 = 5 n.s
Inst 3 = 5 n.s
Inst 4 = 5 n.s
Inst 5 = 5 n.s

Because clock cycle Time = 5 n.s one Inst will take one clock cycle.

Total Time in Single cycle Arch = 5+5+5+5+5
= 25 n.s

## Draw Backs:

i → One In and one out point.

ii → If one Inst is in the Architecture, other all will be in wait state.

## Solution:-

Total Time = 4+3+4+5+3 = 19 n.s

Inst 1 [4 n.s]
Inst 2 [3 n.s]
Inst 3 [4 n.s]
Inst 4 [5 n.s]
Inst 5 [3 n.s]

Multi clock cycle Arch.

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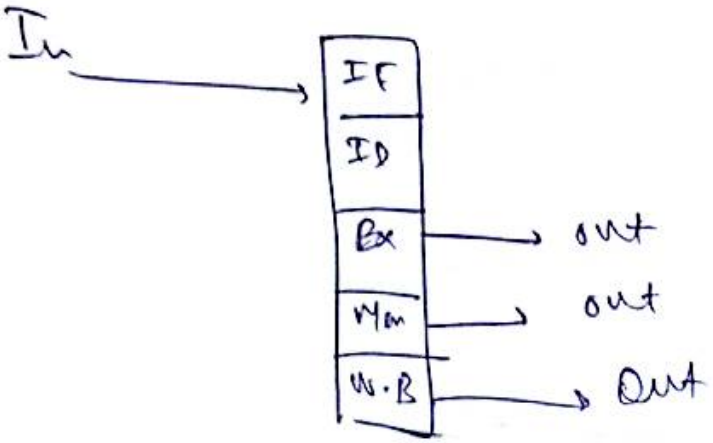
Comparison.

| Single cycle Arch | Multi cycle Arch |
|---|---|
| → One clock time for all Inst. (Arch Same) | → Different clock time for different type of Instr. |
| → One in and one out point | → One in but Multi out points. |




## Draw backs

→ If one Instruction is in The Architecture, then no other instruction will be fetched.

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(6) Example

clothes collection → washing → Spinning → Folding

Single cycle: one drum → wash → spin → fold → pack

Multi cycle: 1 cloth → wash

2 cloth → wash → Spin

3 cloth → wash → spin → fold.

⇒ Pipelining:

I.F → I.D → Exec → Mem. → W.B

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 n.s |
|---|---|---|---|---|---|---|---|---|---|
| (1) | I.F | I.D | Exec | M | W.B | | | | |
| (2) | | IF | ID | Exec | Mem. | W.B | | | |
| (3) | | | IF | ID | Exec | Mem. | W.B | | |
| (4) | | | | IF | ID | Exec | M. | W.B | |
| (5) | | | | | IF | ID | Exec | M | W.B |

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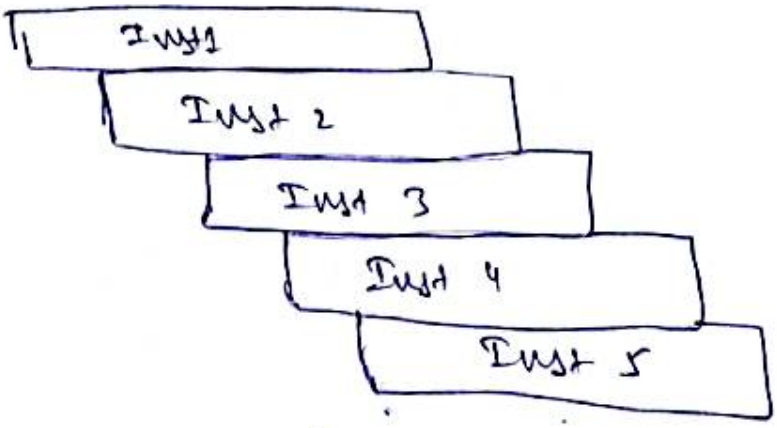
7 Comparison.

| Single cycle | Multi cycle | Pipelining |
|---|---|---|
| Total Inst = 5 | | |
| Time = 5×5 = 25 n.s | Time = 19 n.s | Time = 9 n.s |
| Sequential Operation | | parallel Operation |

## Draw Back

→ Single cycle Pipeline model.




→ Issues → Hazards

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# Hazards (Issues / dependency)

i. Structural Hazards.
→ Solution → Wait State

ii. Data Hazards

iii. Control Hazards

i. <u>Structural</u> (H/W related)

Same H/W → Same time
↑ Req
Different Instruction

<u>Solution</u>: i. Same H/W → Different Time (wait)

ii. Different H/W → Same Time. ✗
↓
practically logical error.

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?

IF — ID — Ex — M — WB

IF — ID — Ex — [M] → W.B

IF — ID — Ex — M — W.B

IF — ID — Ex — M — W.B

[I.F] — ID — Ex — M — W.B

In case of shared Memory.

Wait → Stall → Bubble

IF ID Exec Mem W.B

IF ID Ex [M] W.B

IF ID Ex Mem W.B

IF ID Exec Mem W.B

Structural Hazard of shared Memory.

[IF] ID Ex M W.B

Stall IF ID Ex Mem

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# Data Hazards

- Arithmetic Inst → Result → After Execution — Only Forwarding
- Memory Access Inst. → Result → After Memory — F + Stall

e.g #1

① add $t0, t1, t2$

② Sub $t3, t0, t4$

① IF → ID $\xrightarrow{t1,t2}$ Exec (3 ns) $\xrightarrow{t0}$ Memory (4 ns) — W.B (5 ns update)

Forwarding

② IF — ID $\xrightarrow{t0,t4}$ Exec — M → W.B

e.g #2

① IF → ID → Exec → Mem (4 ns) → W.B (Result update)

IF — ID — Ex — M — W.B

Solution

IF → ID → Exec → M → W.B

F

(Stall) IF — ID — Ex — M — WB

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## Control Hazards

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Conditional Instruction Hazard.

→ ⇒ stalls and the solution

```
if ( a > b)
{
  z = x+y;
}
else
  z = x-y;
```

beq $t0, $t1, Label1

add —
sub —
mul —

beq → IF → ID – EX – M – WB (operation)

add, sub: ID & M – WB

Label 1:

and —
or —
push —
pop —

1. wrong Inst were fetched.
2. Stalls.

IF – ID – EX – M – WB

st st IF ID —

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